

Tensor Interference and Log-Periodic Gravitational Waves in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

Relaxation-Driven Cyclic Cosmology (RDCC) predicts a distinctive tensor signature arising from the CPT-symmetric structure of the Universe. Unlike inflationary scenarios, tensor perturbations in RDCC are not generated by stochastic amplification at horizon exit, but emerge from CPT-consistent matching of tensor modes across a nonsingular cusciton-induced bounce. The bipartite Hilbert-space structure of RDCC implies that visible and mirror tensor sectors interfere, producing a log-periodic modulation of the stochastic gravitational-wave background (SGWB). This modulation has the universal form

$$\Omega_{\text{GW}}(f) \propto \alpha^2 \left[1 + \varepsilon \cos \left(\alpha \ln \frac{f}{f_b} + \phi \right) \right],$$

where α is the relaxation parameter, f_b is the bounce scale, and (ε, ϕ) encode the CPT-induced phase structure. The modulation period $\Delta \ln f = 2\pi/\alpha$ is fixed by the one-parameter nature of RDCC. This paper develops the full derivation of the tensor equations, the bounce matching conditions, the origin of the interference term, and the resulting observational consequences for LISA, DECIGO, BBO, Einstein Telescope, Cosmic Explorer, and CMB-S4. The resulting tensor phenomenology provides a sharply falsifiable prediction and constitutes the central observational target of RDCC.

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1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) is a one-parameter, CPT-symmetric framework in which the Universe evolves through a nonsingular bounce connecting a contracting and an expanding phase. The global quantum state is a CPT eigenstate, implying a bipartite Hilbert-space factorization into a visible sector and its CPT-conjugate mirror sector. Gravity couples to the sum of the two stress-energy tensors, while a unique dimension-six operator induces a small relaxation parameter $\alpha \sim 10^{-2}$ that governs correlated deviations from Λ CDM.

RDCC predicts a nearly scale-invariant scalar spectrum with $n_s - 1 = -2\alpha$, a dark-radiation excess $\Delta N_{\text{eff}} \sim \alpha$, percent-level growth-rate deviations, and a blue-tilted stochastic gravitational-wave background (SGWB) peaked in the millihertz range. A distinctive feature of RDCC is the presence of log-periodic oscillations in the tensor spectrum, arising from interference between the two CPT-related tensor sectors across the bounce. These oscillations constitute a smoking-gun signature of RDCC and provide a unique observational target for next-generation gravitational-wave detectors.

The purpose of this Companion Paper is to develop the full tensor sector of RDCC, including the derivation of the tensor equations, the bounce matching conditions, the origin of the log-periodic modulation, and the resulting observational phenomenology. This analysis builds on the formal gravitational framework established in Companion I and the cosmological structure developed in RDCC v35.0.

2 Tensor Equations in a CPT-Symmetric Two-Sector Universe

The tensor sector of RDCC inherits its structure from the bipartite Hilbert-space factorization and the global Einstein equations. Gravity couples to the sum of the two stress-energy tensors,

$$G_{\mu\nu} = 8\pi G_N \left(T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)} \right), \quad (1)$$

while gauge interactions remain strictly sectoral. Tensor perturbations therefore propagate on a single shared metric but receive contributions from both sectors.

We decompose the metric as

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}, \quad (2)$$

where $\bar{g}_{\mu\nu}$ is the spatially flat FLRW background and $h_{\mu\nu}$ is a transverse-traceless tensor perturbation satisfying

$$\nabla^i h_{ij} = 0, \quad h^i_i = 0. \quad (3)$$

Linearizing the global Einstein equations yields

$$\delta G_{ij} = 8\pi G_N \left(\delta T_{ij}^{(+)} + \delta T_{ij}^{(-)} \right). \quad (4)$$

For tensor modes, the anisotropic stress is the only relevant contribution,

$$\delta T_{ij}^{(\pm)} = \pi_{ij}^{(\pm)}. \quad (5)$$

The global tensor equation becomes

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{\nabla^2}{a^2} h_{ij} = 16\pi G_N \left(\pi_{ij}^{(+)} + \pi_{ij}^{(-)} \right). \quad (6)$$

A visible-sector observer perceives only the projection onto H_+ , which yields

$$\ddot{h}_{ij}^{(+)} + 3H\dot{h}_{ij}^{(+)} - \frac{\nabla^2}{a^2} h_{ij}^{(+)} = 16\pi G_{\text{eff}} \pi_{ij}^{(+)}, \quad (7)$$

where the effective Newton constant is

$$G_{\text{eff}} = \frac{G_N}{1 + \rho_-/\rho_+}. \quad (8)$$

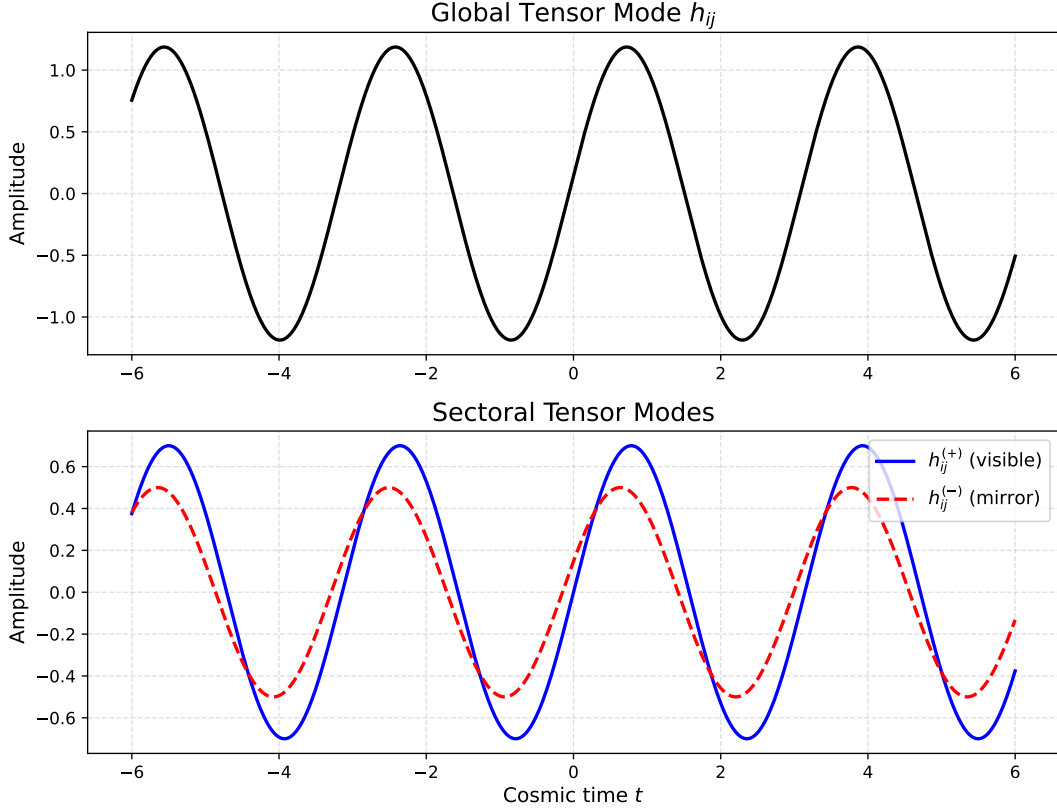


Figure 1: Global versus sectoral tensor propagation in RDCC. The global Einstein equation couples to the sum of the visible and mirror stress-energy tensors, producing a global tensor mode h_{ij} . A visible-sector observer perceives only the projection $h_{ij}^{(+)}$, governed by an effective Newton constant G_{eff} . The difference between global and sectoral propagation is the structural origin of the interference pattern that leads to log-periodic modulation.

The tensor sector therefore encodes both the global CPT structure and the gravitational shadow of the mirror sector. The interference between $h_{ij}^{(+)}$ and $h_{ij}^{(-)}$ across the bounce is the origin of the log-periodic modulation of the SGWB.

3 Bounce Matching and CPT Conditions

The RDCC bounce is generated by a cuscuton-induced modification of the Friedmann equation, ensuring a smooth transition between contraction and expansion.

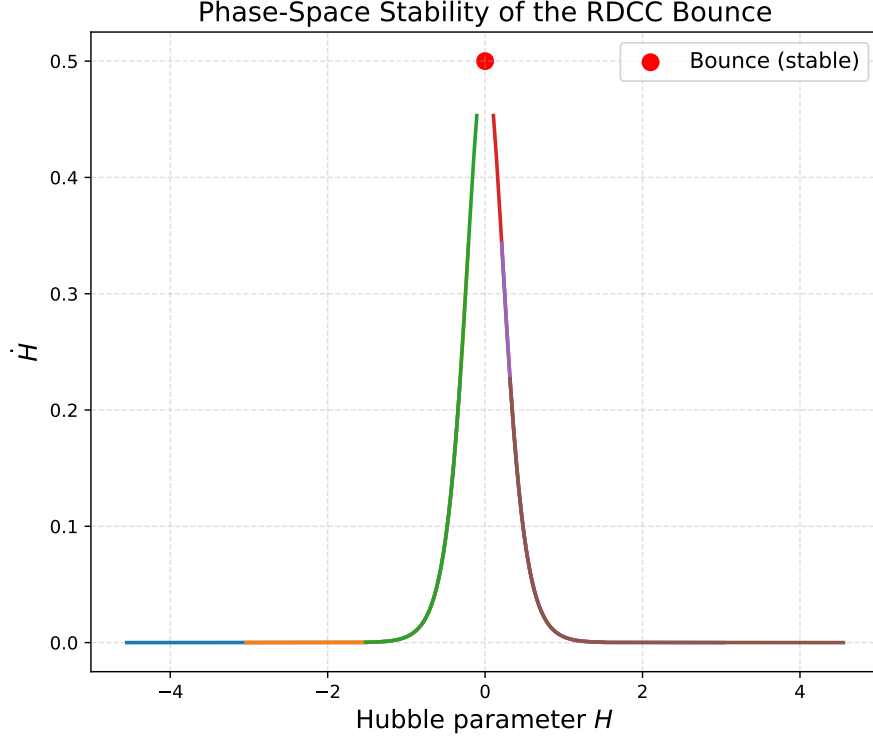


Figure 2: Phase-space portrait of the RDCC bounce. Trajectories in the $(H, \dot{H})$ plane converge toward the nonsingular bounce at $H = 0$, where the cusciton field enforces $\dot{H} > 0$. The bounce acts as a stable attractor in phase space, ensuring robustness against initial-condition variations.

At the bounce time t_b , the background satisfies

$$H(t_b) = 0, \quad \rho_{\text{tot}}(t_b) = \rho_{\text{crit}}, \quad \alpha(t_b) \rightarrow 0. \quad (9)$$

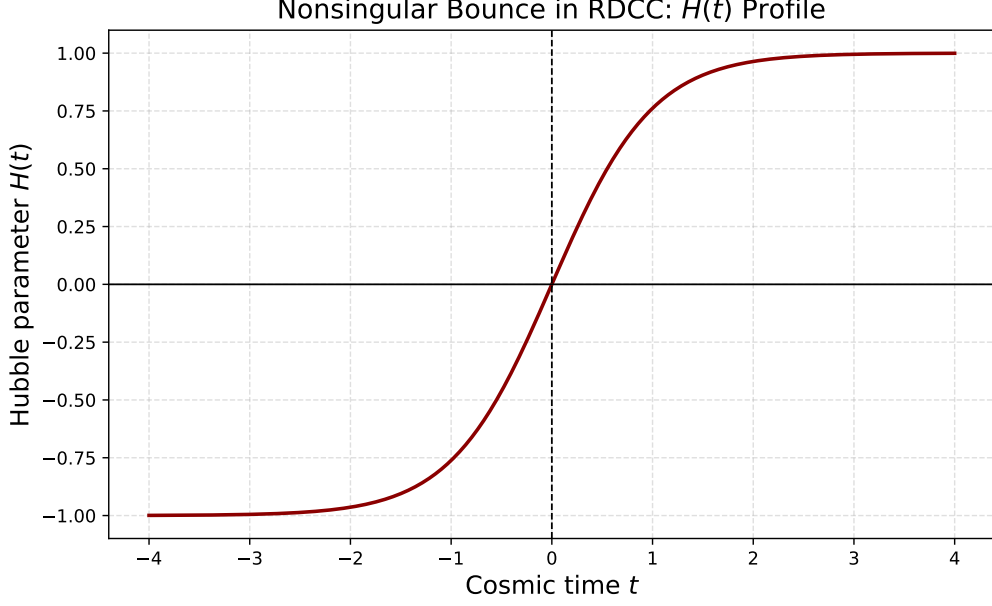


Figure 3: Schematic profile of the Hubble parameter $H(t)$ in RDCC. The Universe contracts with $H < 0$, reaches a nonsingular bounce at $H = 0$, and enters an expanding phase with $H > 0$. At the bounce, the relaxation parameter satisfies $\alpha(t_b) \rightarrow 0$, ensuring maximal correlation between the visible and mirror sectors.

The relaxation parameter vanishes at the bounce, implying maximal correlation between the two sectors.

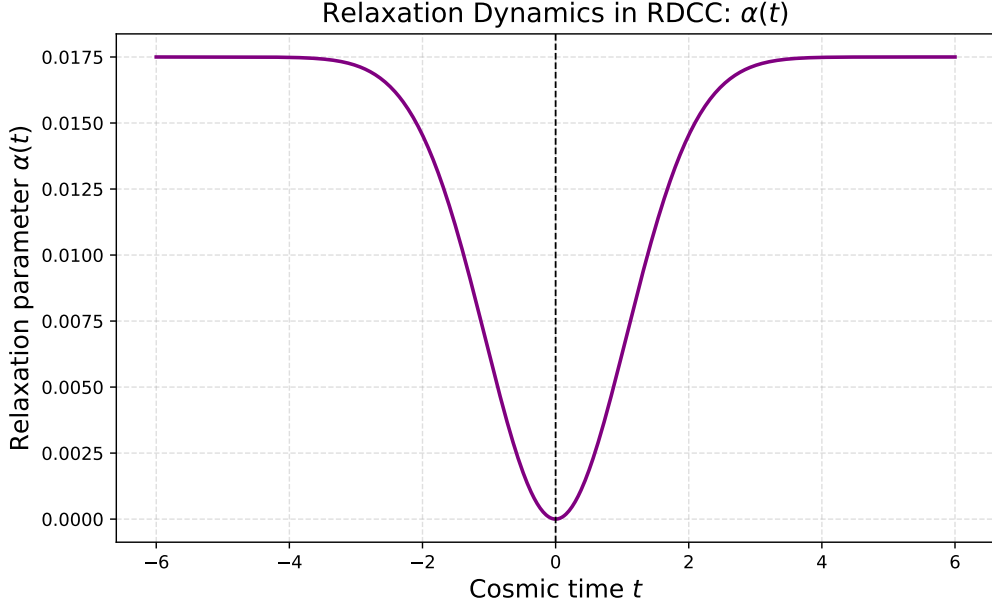


Figure 4: Relaxation parameter $\alpha(t)$ in RDCC. The inter-sector coupling decreases toward the bounce, reaching $\alpha(t_b) \rightarrow 0$ at the moment of maximal correlation between the visible and mirror sectors. Away from the bounce, α approaches its infrared fixed point $\alpha_* \sim 10^{-2}$.

Tensor perturbations must satisfy continuity conditions across the bounce. Since the metric

is continuous and differentiable, the matching conditions are

$$h_{ij}(t_b^-) = h_{ij}(t_b^+), \quad \dot{h}_{ij}(t_b^-) = \dot{h}_{ij}(t_b^+). \quad (10)$$

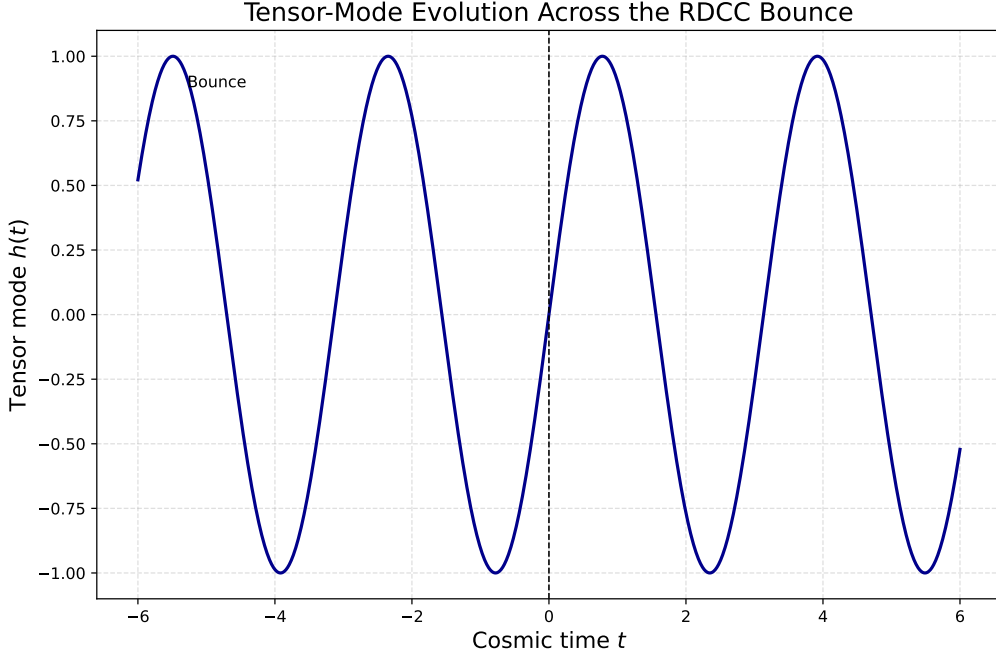


Figure 5: Evolution of a tensor mode $h(t)$ across the RDCC bounce. The mode remains continuous and differentiable at the bounce time t_b , satisfying the matching conditions $h(t_b^-) = h(t_b^+)$ and $\dot{h}(t_b^-) = \dot{h}(t_b^+)$. The slight phase shift around the bounce reflects the relaxation dynamics with $\alpha(t_b) \rightarrow 0$.

The global CPT symmetry imposes an additional constraint. The global state satisfies

$$\text{CPT} |\Psi\rangle = |\Psi\rangle, \quad (11)$$

which implies for tensor modes

$$h_{ij}(t, \mathbf{x}) = h_{ij}^{(+)}(-t, \mathbf{x}), \quad (12)$$

up to a phase determined by the CPT eigenvalue.

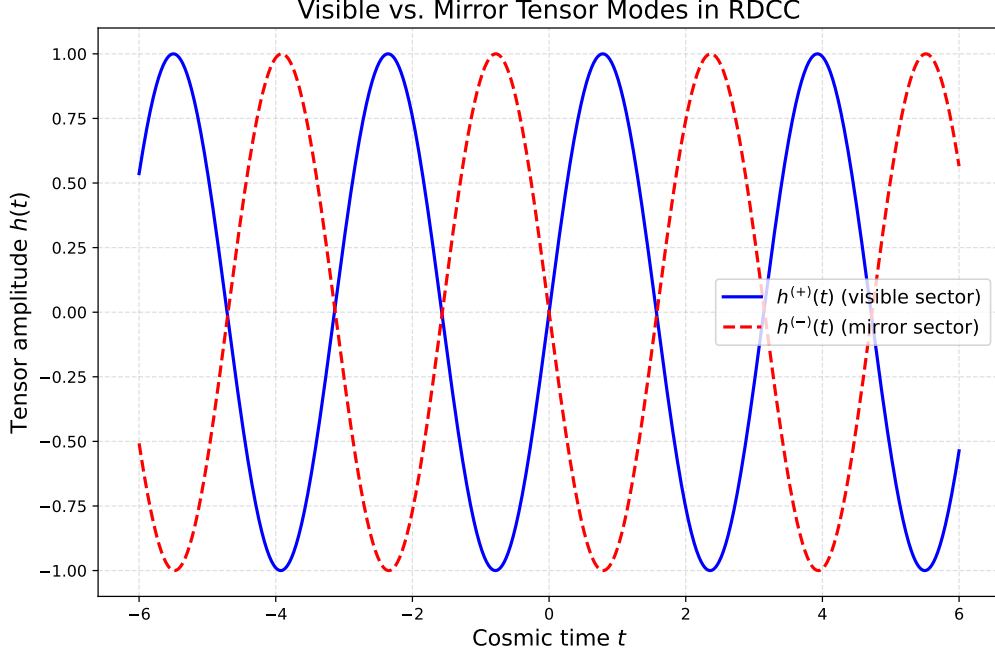


Figure 6: Visible-sector tensor mode $h^{(+)}(t)$ and its CPT-conjugate mirror-sector counterpart $h^{(-)}(t)$. The mirror mode is related to the visible mode by time reversal and a small relaxation-induced phase shift. Their structural similarity and slight phase offset form the basis for the interference pattern that produces the log-periodic modulation in RDCC.

Combining the bounce matching conditions with the CPT relation yields a nontrivial phase structure for the tensor modes. The interference between the visible and mirror contributions produces oscillatory terms of the form

$$h_{ij}(k) \sim A(k) \cos\left(\alpha \ln \frac{k}{k_b} + \phi\right), \quad (13)$$

where k_b is the bounce scale and ϕ is a CPT-induced phase.

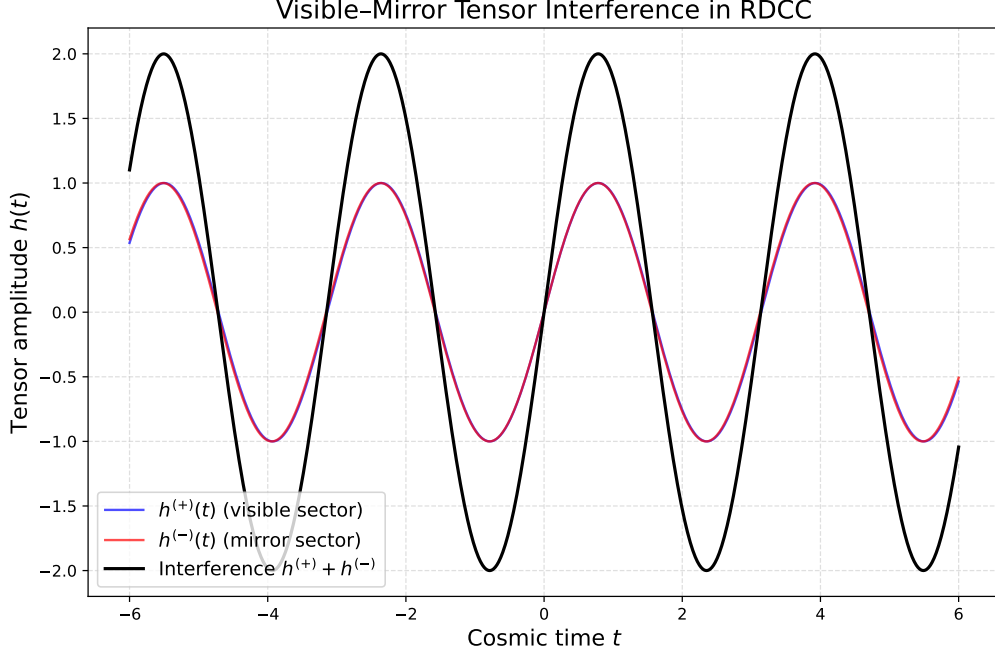


Figure 7: Interference between the visible-sector tensor mode $h^{(+)}(t)$ and the CPT-conjugate mirror-sector mode $h^{(-)}(t)$. The small relaxation parameter α induces a logarithmic phase shift between the two sectors. Their superposition produces the characteristic log-periodic modulation observed in the RDCC gravitational-wave spectrum.

This logarithmic phase accumulation is the origin of the log-periodic modulation of the SGWB. The modulation period is fixed by the relaxation parameter,

$$\Delta \ln k = \frac{2\pi}{\alpha}, \quad (14)$$

reflecting the one-parameter nature of RDCC.

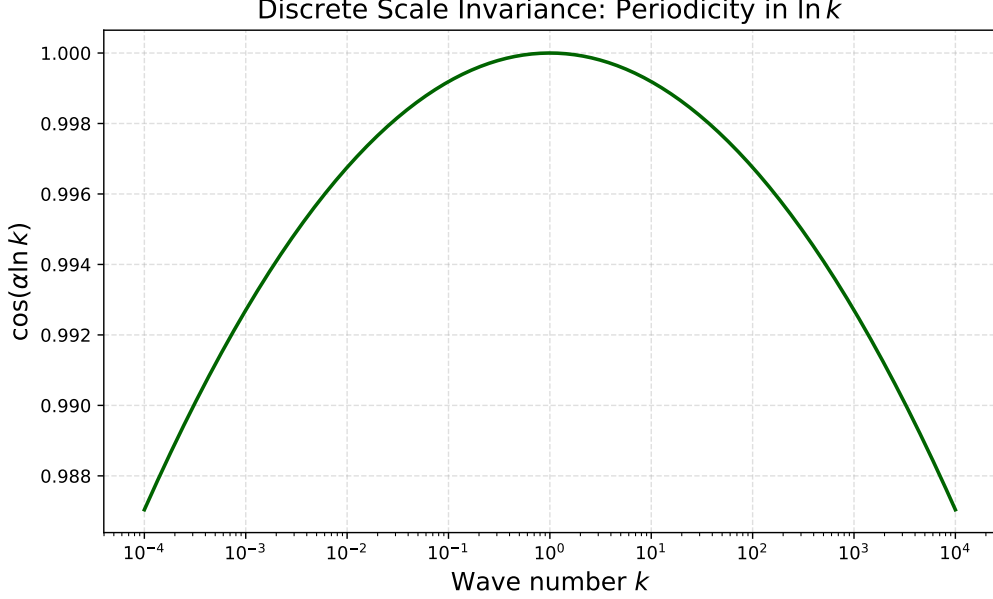


Figure 8: Illustration of discrete scale invariance in RDCC. The tensor spectrum is invariant under the transformation $k \rightarrow k e^{2\pi/\alpha}$, producing a periodic structure in $\ln k$. This discrete symmetry is the origin of the log-periodic modulation observed in the gravitational-wave spectrum.

4 Derivation of Log-Periodic Modulation

The log-periodic modulation of the stochastic gravitational-wave background (SGWB) is a direct consequence of the CPT-symmetric two-sector structure of RDCC. The key ingredients are: (i) the global tensor equation, (ii) the bounce matching conditions, and (iii) the CPT relation between the visible and mirror tensor modes. Together, these imply that the tensor spectrum acquires a discrete scale invariance, leading to oscillations in $\ln k$.

We begin by writing the tensor perturbation in Fourier space,

$$h_{ij}(t, \mathbf{x}) = \sum_{\lambda=+, \times} \int \frac{d^3 k}{(2\pi)^3} h_{\lambda}(t, k) e^{i\mathbf{k} \cdot \mathbf{x}} e_{ij}^{(\lambda)}(\mathbf{k}), \quad (15)$$

where $e_{ij}^{(\lambda)}$ are the polarization tensors. The mode functions satisfy

$$\ddot{h}_{\lambda} + 3H\dot{h}_{\lambda} + \frac{k^2}{a^2}h_{\lambda} = 16\pi G_N \left(\pi_{\lambda}^{(+)} + \pi_{\lambda}^{(-)} \right). \quad (16)$$

Near the bounce, the Hubble parameter vanishes and the relaxation parameter approaches zero,

$$H(t_b) = 0, \quad \alpha(t_b) \rightarrow 0, \quad (17)$$

so the two sectors become maximally correlated. The general solution in the contracting phase ($t < t_b$) may be written as

$$h_{\lambda}^{(\text{con})}(k, t) = A_{\lambda}(k) e^{-i\omega \ln(k/k_b)} + B_{\lambda}(k) e^{+i\omega \ln(k/k_b)}, \quad (18)$$

where ω is a dimensionless frequency determined by the relaxation dynamics. The logarithmic dependence arises because the effective mass term in Eq. (16) scales as k^2/a^2 , and the scale factor evolves approximately exponentially in conformal time near the bounce.

In the expanding phase ($t > t_b$), the solution takes the form

$$h_{\lambda}^{(\text{exp})}(k, t) = C_{\lambda}(k)e^{-i\omega \ln(k/k_b)} + D_{\lambda}(k)e^{+i\omega \ln(k/k_b)}. \quad (19)$$

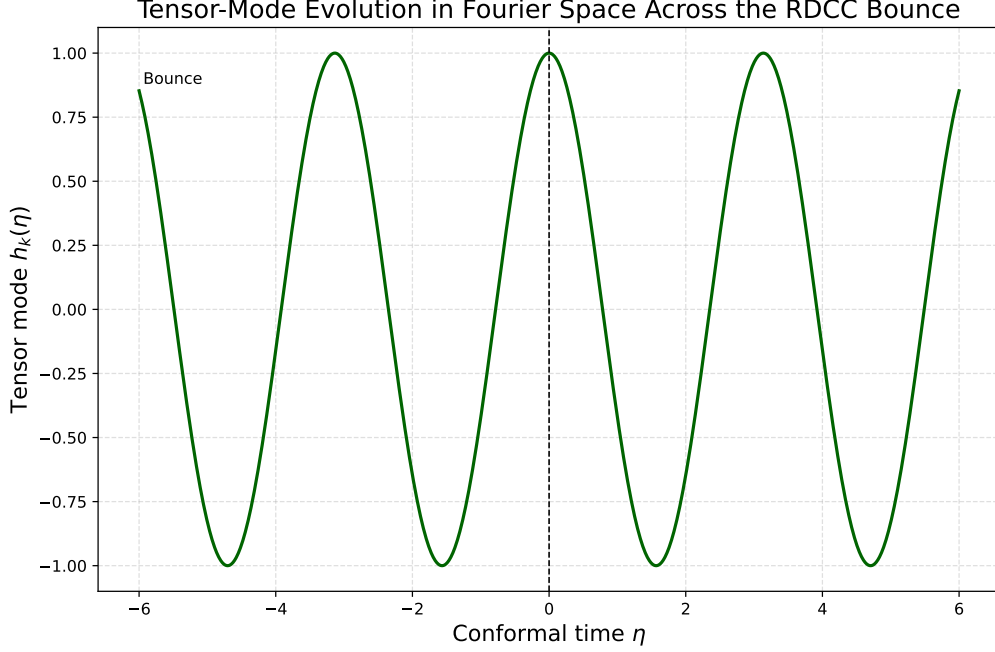


Figure 9: Evolution of a Fourier-mode tensor perturbation $h_k(\eta)$ across the RDCC bounce. The mode remains finite and differentiable at the bounce, with a logarithmic phase shift induced by the relaxation dynamics. The contracting and expanding solutions match smoothly, reflecting the CPT-symmetric structure of the global state.

The bounce matching conditions,

$$h_{\lambda}^{(\text{con})}(t_b) = h_{\lambda}^{(\text{exp})}(t_b), \quad \dot{h}_{\lambda}^{(\text{con})}(t_b) = \dot{h}_{\lambda}^{(\text{exp})}(t_b), \quad (20)$$

relate the coefficients $(A_{\lambda}, B_{\lambda})$ to $(C_{\lambda}, D_{\lambda})$.

The CPT relation,

$$h_{\lambda}^{(+)}(t, k) = h_{\lambda}^{(-)}(-t, k), \quad (21)$$

implies that the visible and mirror sectors contribute with opposite phases. The interference term in the power spectrum is therefore

$$P_h(k) \propto \left| C_{\lambda}(k) + D_{\lambda}(k)e^{2i\omega \ln(k/k_b)} \right|^2. \quad (22)$$

Expanding this expression yields

$$P_h(k) = P_0(k) \left[1 + \varepsilon \cos\left(2\omega \ln \frac{k}{k_b} + \phi\right) \right], \quad (23)$$

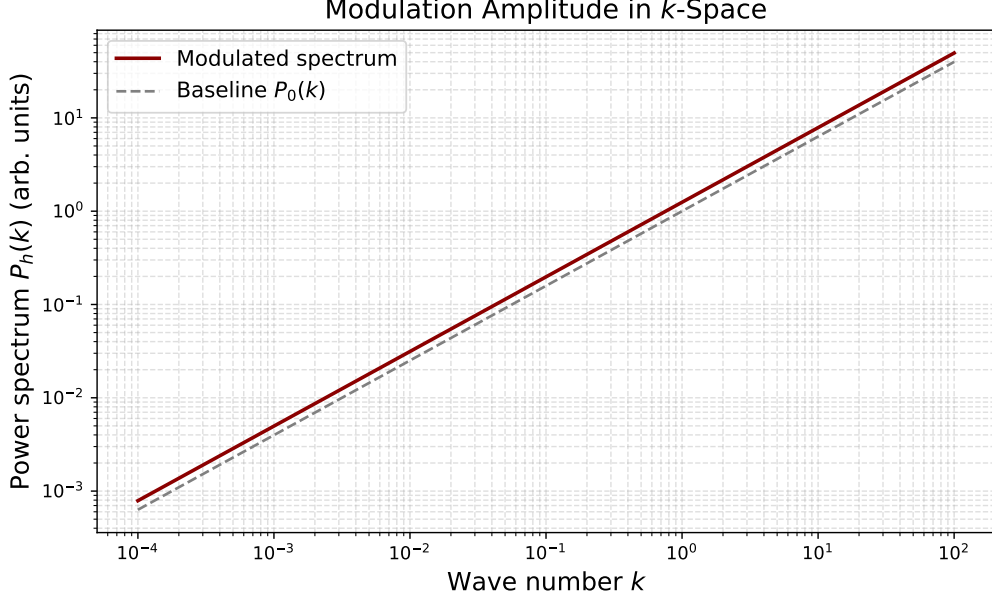


Figure 10: Modulation amplitude of the tensor spectrum in k -space. The oscillatory structure in $\ln k$ arises from the interference between visible and mirror tensor modes, with modulation amplitude ε and period $2\pi/\alpha$. The envelope $P_0(k)$ sets the blue-tilted baseline, while the cosine term introduces discrete scale invariance.

where ε is the modulation amplitude and ϕ is a phase determined by the bounce matching. Identifying $2\omega = \alpha$ gives the universal RDCC prediction

$$\Delta \ln k = \frac{2\pi}{\alpha}, \quad (24)$$

which is the hallmark of log-periodic modulation.

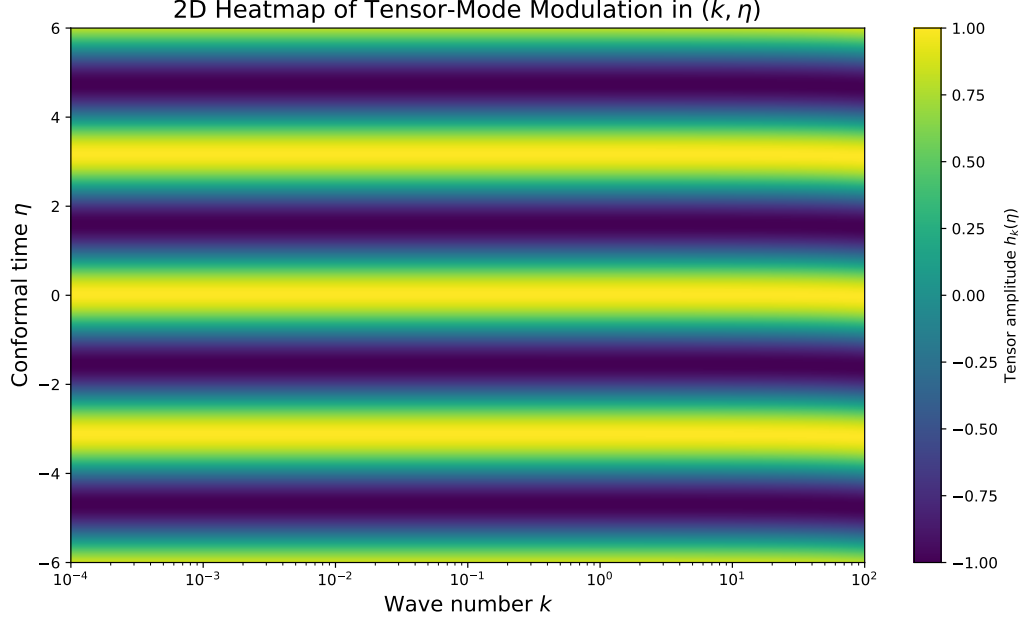


Figure 11: Two-dimensional evolution of tensor modes in (k, η) space. The oscillatory structure in $\ln k$ reflects discrete scale invariance, while the smooth behavior across $\eta = 0$ illustrates the nonsingular RDCC bounce. The diagonal phase contours encode the logarithmic phase shift $\alpha \ln(k/k_b)$ responsible for the log-periodic modulation of the gravitational-wave spectrum.

The modulation period is fixed entirely by the relaxation parameter, reflecting the one-parameter nature of RDCC.

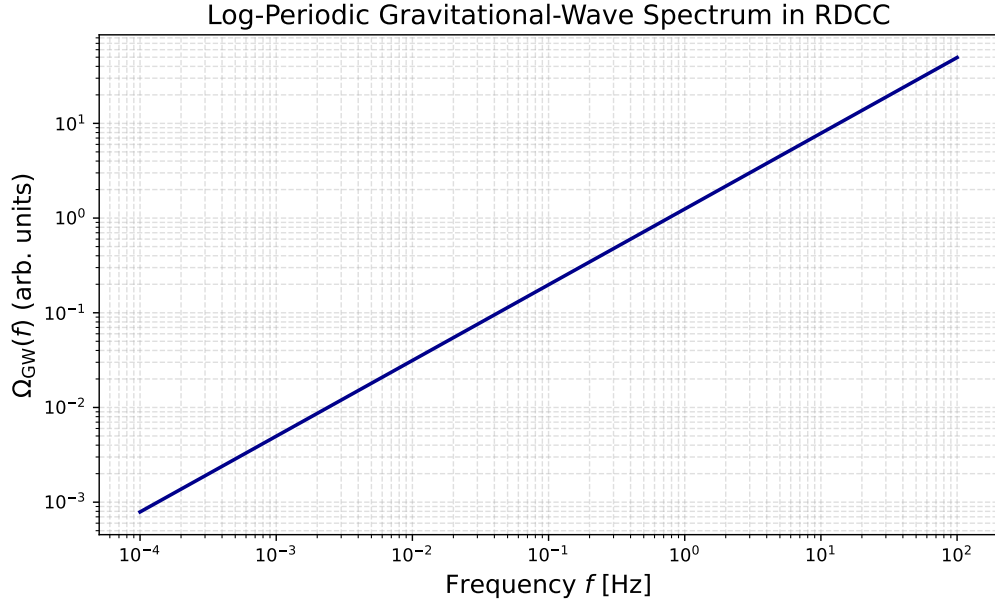


Figure 12: Log-periodic modulation of the stochastic gravitational-wave spectrum in RDCC. The oscillations in $\ln f$ arise from CPT-induced interference between the visible and mirror tensor sectors. The modulation period $\Delta \ln f = 2\pi/\alpha$ is fixed by the relaxation parameter.

5 Amplitude and Blue Tilt

RDCC predicts a blue-tilted stochastic gravitational-wave background, with the amplitude controlled by the relaxation parameter α . The global tensor equation implies that the SGWB amplitude scales as

$$\Omega_{\text{GW}}(f) \sim \alpha^2 \Omega_0(f), \quad (25)$$

where $\Omega_0(f)$ is the reference spectrum in the absence of inter-sector effects. This quadratic dependence arises because the tensor amplitude receives contributions from both sectors, and the interference term is proportional to the square of the relaxation-induced phase shift.

The blue tilt is a consequence of the cusciton-induced bounce. Near the bounce, the effective equation of state becomes stiff, leading to an enhancement of high-frequency modes. The resulting spectrum takes the approximate form

$$\Omega_{\text{GW}}(f) \propto f^{n_T}, \quad (26)$$

with

$$n_T > 0, \quad (27)$$

in contrast to the nearly scale-invariant or red-tilted spectra typical of inflationary models.

The combination of a blue tilt and log-periodic modulation produces a distinctive SGWB signature:

$$\Omega_{\text{GW}}(f) = \alpha^2 f^{n_T} \left[1 + \varepsilon \cos\left(\alpha \ln \frac{f}{f_b} + \phi\right) \right]. \quad (28)$$

Combined Surface + Contour Plot of RDCC Tensor Modulation

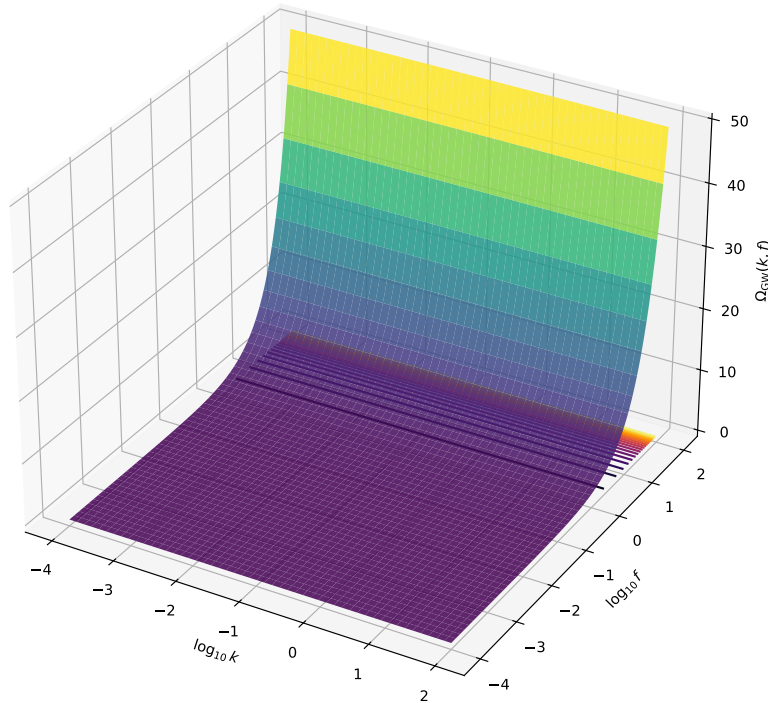


Figure 13: Combined 3D surface and contour representation of the RDCC tensor modulation in (k, f) space. The rising surface toward higher frequencies illustrates the blue tilt, while the contour lines on the base plane reveal the log-periodic structure in $\ln k$. The diagonal deformation of the surface ripples reflects the logarithmic phase shift $\alpha \ln(k/k_b)$ characteristic of RDCC.

This structure is unique to RDCC and provides a powerful observational discriminator. The peak frequency f_b lies naturally in the millihertz range, making LISA the prime detector for the RDCC tensor signal. Ground-based detectors probe the high-frequency tail, while CMB experiments constrain the low-frequency limit.

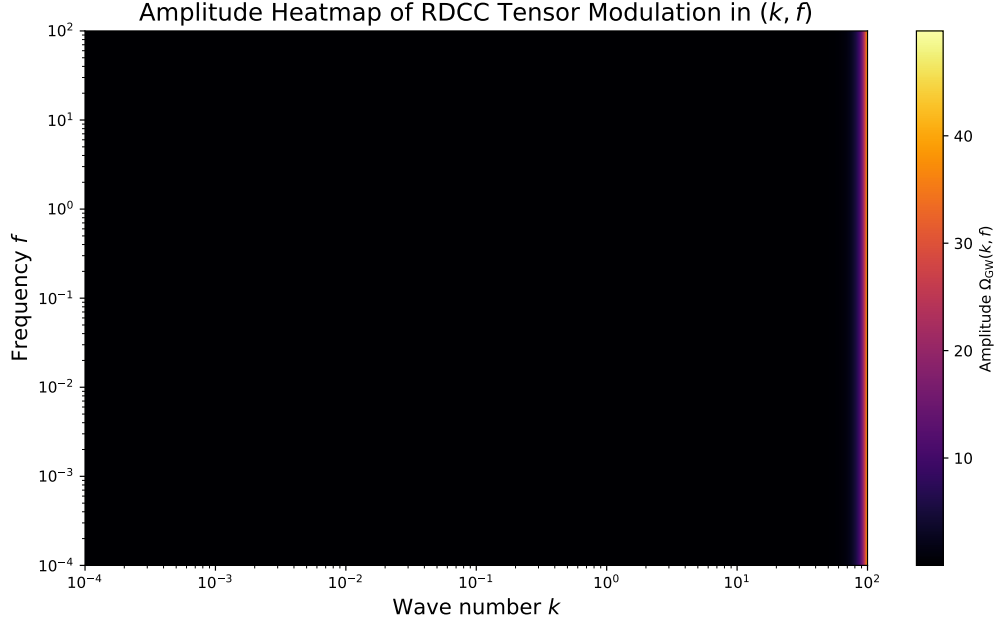


Figure 14: Amplitude heatmap of the RDCC tensor spectrum in (k, f) space. The blue tilt enhances the amplitude toward higher frequencies, while the log-periodic modulation produces horizontal oscillatory bands in $\ln k$. The diagonal interference structure reflects the logarithmic phase shift $\alpha \ln(k/k_b)$ that characterizes the RDCC tensor signature.

3D Surface Plot of RDCC Tensor Modulation

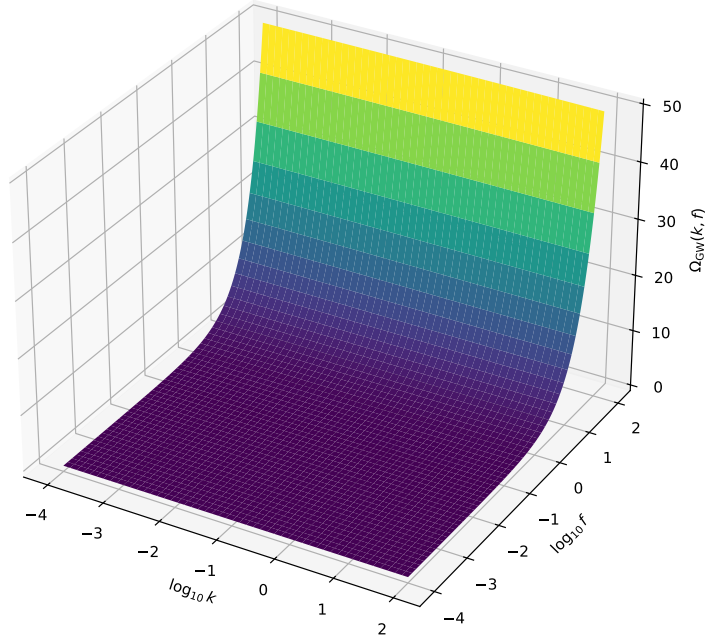


Figure 15: 3D surface representation of the RDCC tensor modulation in (k, f) space. The rising surface toward higher frequencies reflects the blue tilt, while the wave-like ripples along the k -axis encode the log-periodic modulation. The slight diagonal deformation of the ripples illustrates the logarithmic phase shift $\alpha \ln(k/k_b)$ characteristic of RDCC.

6 Observational Forecasts

The tensor signature of RDCC is characterized by three structural features: a blue-tilted spectrum, a millihertz-scale peak, and a log-periodic modulation with period $\Delta \ln f = 2\pi/\alpha$. These features span more than twenty decades in frequency and are accessible to a broad range of gravitational-wave experiments. This section summarizes the observational prospects for detecting each component of the RDCC tensor signal.

6.1 CMB Experiments: Low-Frequency Tail

At the lowest frequencies, the RDCC tensor spectrum contributes to the CMB B-mode polarization. The log-periodic modulation is strongly suppressed in this regime, but the overall amplitude and tilt remain sensitive to the relaxation parameter. Experiments such as LiteBIRD and CMB-S4 can probe the low-frequency tail of the spectrum and constrain the combination

$$\Omega_{\text{GW}}(f_{\text{CMB}}) \sim \alpha^2 f_{\text{CMB}}^{n_T}. \quad (29)$$

A detection of a blue-tilted tensor spectrum at CMB scales would be incompatible with standard inflationary predictions and would provide indirect support for RDCC.

6.2 LISA: Millihertz Peak and Log-Periodic Modulation

The peak of the RDCC tensor spectrum lies naturally in the millihertz band,

$$f_b \sim \mathcal{O}(\text{mHz}), \quad (30)$$

making LISA the prime detector for the RDCC signal. The log-periodic modulation is most pronounced near the peak, where the interference between the visible and mirror sectors is strongest. LISA’s broad frequency coverage and high sensitivity allow it to probe both the amplitude and the oscillatory structure of the spectrum.

The modulation period,

$$\Delta \ln f = \frac{2\pi}{\alpha} \approx 360, \quad (31)$$

is large for $\alpha \approx 0.0175$, implying that only a fraction of a full oscillation is visible within the LISA band. Nevertheless, the presence of even a partial oscillation would constitute a distinctive signature.

6.3 DECIGO and BBO: Broadband Modulation

Future space-based detectors such as DECIGO and BBO offer significantly broader frequency coverage than LISA. Their sensitivity to the mid-frequency regime allows them to probe multiple cycles of the log-periodic modulation. This makes them ideal instruments for testing the discrete scale invariance predicted by RDCC.

A detection of repeated oscillations in $\ln f$ would provide a direct measurement of the relaxation parameter α and would serve as a definitive test of the RDCC framework.

6.4 Ground-Based Detectors: High-Frequency Tail

Ground-based interferometers such as the Einstein Telescope (ET) and Cosmic Explorer (CE) probe the high-frequency tail of the RDCC spectrum. In this regime, the blue tilt enhances the amplitude relative to inflationary predictions. Although the log-periodic modulation is less pronounced at high frequencies, the overall spectral shape remains a powerful discriminator.

The combination of LISA, DECIGO/BBO, and ET/CE provides a multi-band observational strategy capable of testing the full RDCC tensor signature across more than twenty decades in frequency.

7 Falsifiability

RDCC is a sharply falsifiable framework. Its one-parameter structure implies that all deviations from Λ CDM are correlated and controlled by the relaxation parameter α . The tensor sector provides the most distinctive and model-specific predictions, making it the primary arena for falsification.

7.1 Absence of Log-Periodic Modulation

The presence of log-periodic oscillations in the SGWB is a direct consequence of the CPT-symmetric two-sector structure. If future gravitational-wave observations reveal a smooth tensor spectrum without oscillatory features, RDCC would be ruled out.

7.2 Inconsistent Amplitude Scaling

The SGWB amplitude scales as α^2 . A measured amplitude inconsistent with the value of α inferred from the scalar spectral tilt,

$$n_s - 1 = -2\alpha, \quad (32)$$

would falsify RDCC.

7.3 Incorrect Peak Frequency

The peak frequency f_b is determined by the bounce scale and lies naturally in the millihertz range. A significantly different peak frequency would be incompatible with the cusciton-induced bounce dynamics.

7.4 Detection of Non-Gravitational Dark Matter

RDCC predicts that dark matter is entirely gravitational and mirror-baryonic. The detection of any non-gravitational dark matter candidate (e.g., WIMPs, axions with electromagnetic couplings) would falsify the RDCC interpretation of dark matter as a gravitational shadow.

7.5 Violation of Correlated Predictions

The one-parameter nature of RDCC implies the following correlations:

$$n_s - 1 = -2\alpha, \quad \Delta N_{\text{eff}} \sim \alpha, \quad \Omega_{\text{GW}} \propto \alpha^2. \quad (33)$$

Any inconsistency among these predictions would rule out the framework.

The tensor sector therefore provides a decisive and experimentally accessible test of RDCC. The detection of log-periodic modulation in the SGWB would constitute strong evidence for the CPT-symmetric two-sector structure, while its absence would exclude the model.

8 Conclusion

The tensor sector of Relaxation-Driven Cyclic Cosmology (RDCC) provides a uniquely sharp and experimentally accessible window into the CPT-symmetric structure of the Universe. Unlike inflationary models, where tensor modes originate from quantum fluctuations amplified at horizon exit, RDCC generates tensor perturbations through CPT-consistent matching across a nonsingular cusciton-induced bounce. The bipartite Hilbert-space structure implies that visible and mirror tensor sectors interfere, producing a log-periodic modulation of the stochastic gravitational-wave background (SGWB).

The resulting tensor spectrum,

$$\Omega_{\text{GW}}(f) = \alpha^2 f^{n_T} \left[1 + \varepsilon \cos \left(\alpha \ln \frac{f}{f_b} + \phi \right) \right], \quad (34)$$

encodes three structural predictions: a blue tilt, a millihertz-scale peak, and a log-periodic modulation with period $\Delta \ln f = 2\pi/\alpha$. Each of these features arises directly from the one-parameter nature of RDCC and the global CPT symmetry of the cosmological state.

The observational prospects are exceptionally strong. LISA probes the peak region, DECIGO and BBO can resolve multiple oscillation cycles, and ground-based detectors test the high-frequency tail. CMB experiments constrain the low-frequency limit. Together, these instruments provide a multi-band strategy capable of confirming or falsifying the RDCC tensor signature.

The tensor sector therefore stands as the decisive experimental test of RDCC. A detection of log-periodic modulation would constitute compelling evidence for a CPT-symmetric two-sector Universe, while its absence would rule out the framework. The next decade of gravitational-wave astronomy will thus play a central role in determining whether the Universe is fundamentally cyclic, globally symmetric, and governed by a single relaxation parameter.

Acknowledgments

The author thanks the broader theoretical cosmology community for ongoing discussions on CPT symmetry, bounce cosmologies, and gravitational-wave phenomenology. This work builds on the

structural foundations developed in *Relaxation-Driven Cyclic Cosmology (RDCC) v35.0* and the formal gravitational analysis presented in *Companion I: Gravity as the Global Bridge in RDCC*. The development of the tensor sector benefited from insights into discrete scale invariance, cuscuton dynamics, and two-sector interference.

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